ADVANCED OLYMPIAD SERIES

Olympiad Geometry: Worked Solutions

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Research edition ▪ 6 November 2026

Definitions. Certificates. Complete arguments.

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Reading this volume

Sixteen selected problems supply the checked core of this research edition. They are chosen to expose reusable mechanisms, not advertised as a statistically established list of the hardest problems. All are IMO shortlist problems; G10 also appeared as IMO 2017 Problem 4. The source archive contains many additional contests and team-selection records, with their verification status stated separately.

The notation (ABC) means the circumcircle of three noncollinear points. Unless ordinary interior angles are explicitly used, angle equalities are modulo 180°. A second intersection and the circle it defines are assumed to exist as in the problem; coincident or tangent limiting configurations are not silently substituted for a stated distinct intersection. The diagrams in the companion volume are schematic illustrations of proved mechanisms, not additional hypotheses.

G01. Two similarities sharing a secant

Source: IMO shortlist 2015, G1. **Record:** `imo_2015_29f541`.

Problem. In an acute triangle ABC with orthocenter H, complete the parallelogram $ABGH$. On line GH choose I so that AC passes through the midpoint of HI. Let $J \neq C$ be the other intersection of AC with (GCI). Prove $IJ = AH$.

Solution. Write $M = AC \cap GH$, so $MI = MH$. Since $BG \parallel AH \perp BC$ and $GH \parallel AB \perp CH$, the quadrilateral $BGCH$ is cyclic. Hence

$$\angle CGH = \angle CBH = \angle HAC.$$

Together with $BG \parallel AH$ and $GH \parallel AB$, this shows that the right triangles CMH and CGB are similar: their right angles are at H and B, and the angle between MC and MH equals that between GC and GB. Therefore $MH/MC = GB/GC$.

The secants MCJ and MIG of (GCI) give similar triangles MIJ and MCG, so $IJ/CG = MI/MC$. Combining the two ratios yields

$$\frac{IJ}{CG} = \frac{MI}{MC} = \frac{MH}{MC} = \frac{GB}{GC} = \frac{AH}{CG}.$$

Thus $IJ = AH$.

Certificate. The same intermediate ratio connects a secant similarity to an orthocenter similarity.

G02. Reflect the two lines

Source: IMO shortlist 2015, G2. **Record:** `imo_2015_7c01e3`.

Problem. Triangle ABC lies on a circle Ω with center O. A circle centered at A cuts BC at D, E in the order B, D, E, C, and cuts the arcs AB and AC of Ω not containing the third vertex at F, G, respectively. Let $K \neq B$ be the intersection of AB with (BDF), and $L \neq C$ the intersection of AC with (CEG). If distinct lines FK, GL meet at X, prove A, X, O collinear.

Solution. The common chord FG of the two original circles is perpendicular to their line of centers AO, so reflection in AO exchanges F and G. We prove that it also exchanges lines FK and GL. Using directed angles, the three circles give

$$\begin{aligned} \angle KFA &= \angle DFG + \angle GFA - \angle DFK \\ &= \angle CEG + \angle GBA - \angle DBK \\ &= \angle CEG - \angle CBG \\ &= \angle CLG - \angle CAG = \angle AGL. \end{aligned}$$

In the second line, D, E, B, C are collinear; the circle centered at A contains D, E, F, G, while (BDF) contains K. The last line uses (CEG) and Ω. The angle equality is exactly the reflection criterion. Since the two lines are distinct, their intersection is fixed by the reflection and hence lies on AO.

Certificate. A concurrency conclusion on a symmetry axis can be proved by reflecting whole lines.

G03. An angle condition is a distance locus

Source: IMO shortlist 2006, G1. **Record:** `imo_2006_e64448`.

Problem. Let I be the incenter of ABC. An interior point P satisfies $\angle PBA + \angle PCA = \angle PBC + \angle PCB$. Prove $AP \geq AI$, with equality only at $P = I$.

Solution. The four angles in the hypothesis sum to $B + C$. Thus $\angle PBC + \angle PCB = (B + C)/2$ and $\angle BPC = 90^\circ + A/2 = \angle BIC$. The points B, C, I, P lie on one circle.

Let T be the midpoint of the arc BC not containing A on (ABC). Then A, I, T are collinear in that order, and $TB = TC = TI$. For completeness, the first equality follows from the equal arcs, while in triangle BIT the angles at B and I both equal $(A + B)/2$: use $\angle CBT = A/2$ and the incenter angle $\angle BIA = 90^\circ + C/2$. Thus T is the center of (BCI).

The triangle inequality gives $AP + PT \geq AT = AI + IT$. As $PT = IT$, we obtain $AP \geq AI$. Equality requires P to lie on segment AT. The circle has two intersections with this line; the nearer one to A is I, and the other lies beyond T, outside the triangle. Therefore equality holds precisely at I.

Certificate. Identify the fixed-angle circle, then locate its center. An optimization becomes a one-line distance inequality.

G04. Reflection completes a cyclicity proof

Source: IMO shortlist 2012, G2. **Record:** `imo_2012_758467`.

Problem. In cyclic quadrilateral $ABCD$, the diagonals meet at E. The extensions of AD, BC beyond A, B meet at F. Complete parallelogram $ECGD$, and reflect E in AD to obtain H. Prove D, H, F, G concyclic.

Solution. Cyclicity gives similarities $EAB \sim EDC$ and $FAB \sim FCD$. Since $GD = EC$ and $GD \parallel EC$, we get

$$\frac{GD}{EB} = \frac{EC}{EB} = \frac{CD}{AB} = \frac{FD}{FB}, \qquad \angle FDG = \angle FBE.$$

Thus $FDG \sim FBE$, and $\angle FGD = \angle FEB$. In the ordinary interior angles of this configuration, reflection in $FD = AD$ gives $\angle FHD = \angle FED$. Since B, E, D are collinear, $\angle FED = 180^\circ - \angle FEB$. Consequently $\angle FHD + \angle FGD = 180^\circ$, proving the required cyclicity.

Certificate. First obtain the angle needed at the unreflected point; reflection supplies the sign change automatically.

G05. Reflected lines meet on the mirror

Source: IMO shortlist 2013, G2. **Record:** `imo_2013_22bd4a`.

Problem. Let M, N be the midpoints of AB, AC, and T the midpoint of the arc BC not containing A on (ABC). The circle (AMT) meets the perpendicular bisector of AC at an interior point X of ABC; similarly (ANT) meets the perpendicular bisector of AB at an interior point Y. If $K = MN \cap XY$, prove $KA = KT$.

Solution. We prove that reflection in the perpendicular bisector of AT sends M, N to X, Y. Use coordinates with $A = (0, 0)$, the internal angle bisector as the positive horizontal axis, and

$$B = 2u(c, s), \quad C = 2v(c, -s), \quad c = \cos(A/2) > 0, \ s = \sin(A/2) > 0.$$

Here $u, v > 0$, $M = (uc, us)$ and $N = (vc, -vs)$. Writing the equation of (ABC) shows that its other horizontal intercept is $T = (t, 0)$, where $t = (u + v)/c$.

The circle (AMT) has equation $x^2+y^2-tx+(v/s)y = 0$. The perpendicular bisector of AC is $cx-sy = v$. Its two intersections with the circle are

$$X_0 = (t - uc, us), \qquad X_1 = (0, -v/s).$$

These are checked directly in both equations. The second point is outside the angle BAC, so the specified interior point is $X = X_0$. Similarly $Y = (t - vc, -vs)$. Thus $(x, y) \mapsto (t - x, y)$ exchanges M, X and N, Y. It exchanges the two lines MN and XY; their unique intersection lies on its fixed line $x = t/2$, the perpendicular bisector of AT. Hence $KA = KT$.

Certificate. A short coordinate calculation establishes the intended reflection and resolves which of two circle intersections is meant.

G06. Double the cevians

Source: IMO shortlist 2014, G1. **Record:** `imo_2014_a0b822`.

Problem. In an acute triangle, choose P, Q on BC with $\angle PAB = C$ and $\angle QAC = B$. Extend AP, AQ to M, N so that $PM = PA$ and $QN = QA$. Show that BM and CN meet on (ABC).

Solution. The triangles ABP and CAQ are similar, so

$$\frac{BP}{PM} = \frac{BP}{PA} = \frac{AQ}{QC} = \frac{NQ}{QC}.$$

Also $\angle BPM = B + C = \angle NQC$. Thus triangles BPM and NQC are similar by two sides and the included angle. Their remaining angles show that the intersection S of BM, CN satisfies $\angle BSC = B + C = 180^\circ - A$. Equivalently, using directed angles, $\angle BSC = \angle BAC \pmod{180^\circ}$. Hence A, B, S, C are concyclic. The same angle relation shows the two lines cannot be parallel, since $0 < A < 180^\circ$.

Certificate. The doubling construction preserves the side ratios needed for a second similarity.

G07. Pack two similar triangles

Source: IMO shortlist 2014, G2. **Record:** `imo_2014_32e67b`.

Problem. Points K, L, M on BC, CA, AB have concurrent cevians AK, BL, CM. Prove that two of ALM, BMK, CKL have inradii whose sum is at least the inradius r of ABC.

Solution. First suppose the points are interior. Set $a = BK/KC$, $b = CL/LA$, $c = AM/MB$. Ceva gives $abc = 1$. Around the cyclic list a, b, c there is a transition from a value at least one to a value at most one. Relabel cyclically so $a \geq 1 \geq b$. Then $bc = 1/a \leq 1$ and $ca = 1/b \geq 1$.

Through M, draw a parallel to BC meeting AC at X. The inequality $bc \leq 1$ is equivalent to $AM/AB \leq AL/AC$, so X lies on AL. Thus triangle AMX lies inside ALM and has inradius $(AM/AB)r$. Through M, draw a parallel to AC meeting BC at Y. The inequality $ca \geq 1$ puts Y on BK, so $BMY \subset BMK$ and its inradius is $(MB/AB)r$.

An inscribed disk in a contained triangle is a disk in the containing triangle, whose largest contained disk is its incircle. Consequently the inradii of ALM, BMK have sum at least $(AM + MB)r/AB = r$. Boundary cases, if allowed, follow by continuity of triangle inradii, interpreting a degenerate triangle's inradius as zero.

Certificate. Ceva selects a favorable pair; homothety turns their sizes into complementary fractions.

G08. Convert intersection ratios into half angles

Source: IMO shortlist 2016, G2. **Record:** `imo_2016_022acb`.

Problem. Let I be the incenter, M the midpoint of BC, and D the foot from I to BC. The perpendicular to AI through I meets AB, AC at F, E. Let $X \neq A$ be the other common point of (AEF) and (ABC). Prove that XD and AM meet on (ABC).

Solution. Let $Y \neq A$ be the other intersection of AM with (ABC). For a cyclic quadrilateral $PQRS$, if its diagonals meet at U, the area formula gives

$$\frac{QU}{US} = \frac{PQ \cdot QR}{PS \cdot SR},$$

with directed versions when necessary. Apply this to $ABYC$: since $BM = MC$, $BY/CY = AC/AB$. If $D' = XY \cap BC$, the corresponding formula for $XBYC$ gives

$$\frac{BD'}{D'C} = \frac{XB}{XC}\frac{AC}{AB}.$$

Angles in the two original circles show $XBF \sim XCE$, whence $XB/XC = BF/CE$. Put $\beta = B/2$, $\gamma = C/2$. Because AI bisects A and $EF \perp AI$, we have $FI = EI$. In triangles BFI, CEI, the sine rule gives

$$\frac{BF}{FI} = \frac{\sin\gamma}{\sin\beta}, \qquad \frac{CE}{EI} = \frac{\sin\beta}{\sin\gamma}.$$

Therefore

$$\frac{BD'}{D'C} = \frac{\sin 2\beta}{\sin 2\gamma}\frac{\sin^2\gamma}{\sin^2\beta} = \frac{\tan\gamma}{\tan\beta} = \frac{BD}{DC}.$$

The directed ratio is positive, so D' lies inside BC; the ratio uniquely determines it. Hence $D' = D$, giving the required intersection Y.

Certificate. A chord-intersection ratio, one similarity, and a half-angle identity form a closed chain. No mixtilinear-circle facts are needed.

G09. The hidden orthocenter of a pentagon

Source: IMO shortlist 2017, G1. **Record:** `imo_2017_e0103e`.

Problem. A convex pentagon $ABCDE$ satisfies $AB = BC = CD$, $\angle EAB = \angle BCD$, and $\angle EDC = \angle CBA$. Prove that AC, BD and the perpendicular from E to BC concur.

Solution. Write the internal angles of the pentagon as A, B, C, D, E, so $A = C$ and $D = B$. Let I be the intersection of the internal angle bisectors at B, C. Since $AB = BC$ and $BC = CD$, these bisectors are also the perpendicular bisectors of AC, BD. Thus $IA = IC$, $IB = ID$, $BI \perp AC$, and $CI \perp BD$.

The triangles IAB, ICB, ICD are congruent by three sides. Hence $\angle IAB = C/2 = A/2$ and $\angle IDC = B/2 = D/2$. Thus I also lies on the internal bisectors at A, D. Its equal distances to the five side lines, on the interior sides, show that IE is the internal bisector at E as well.

Let $H = AC \cap BD$. The first perpendicular relations say that H is the orthocenter of triangle BIC, so $IH \perp BC$. It remains to put E on IH. The angle sum gives $E = 540^\circ - 2B - 2C$. In quadrilateral $ABIE$,

$$\angle BIE = 360^\circ - C - B/2 - E/2 = 90^\circ + B/2.$$

Since $\angle IBC = B/2$, this means $IE \perp BC$. Consequently E, I, H are collinear, proving the concurrency.

Audit note. The extracted source contains an incorrect plus sign in its displayed pentagon angle sum. The correct sum is used above.

G10. A half-turn creates the tangent

Source: IMO 2017, Problem 4; shortlist G2. **Record:** `imo_2017_bb6ddb`.

Problem. Points R, S lie on Ω, and $R' = 2S - R$. Let I lie on the smaller arc RS. The circle $\Gamma = (ISR')$ meets the tangent to Ω at R twice; call the nearer intersection to R point A. If AI meets Ω again at J, prove JR' tangent to Γ.

Solution. Let $A' = 2S - A$. Thus $ARA'R'$ is a parallelogram. The two circles and the collinearity A, I, J give

$$\angle JRS = \angle JIS = \angle AR'S.$$

As R, S, R' are collinear, this implies $RJ \parallel AR'$, so R, J, A' are collinear. Since $R'A' \parallel RA$ and RA is tangent to Ω,

$$\angle SR'A' = \angle SRA = \angle SJR.$$

Thus S, J, A', R' lie on a circle. It follows that

$$\angle SR'J = \angle SA'J = \angle SA'R = \angle SAR'.$$

This is the tangent–chord criterion for line JR' at the point R' of Γ, as required. The designated nearer intersection is the one satisfying the stated ordinary configuration; the directed-angle argument avoids imposing an additional diagram order.

Certificate. Reflect the difficult point through the given midpoint. A tangent claim becomes a cyclic quadrilateral.

G11. Two tangents encode a reflection

Source: IMO shortlist 2019, G1. **Record:** `imo_2019_be46cd`.

Problem. A circle through A meets AB, AC again at D, E and crosses BC at F, G in that order. Tangents at F to (BDF) and at G to (CEG) meet at $T \neq A$. Prove $AT \parallel BC$.

Solution. The tangent–chord theorem and the first circle give, in directed form,

$$\angle(TF, BC) = \angle(BC, GA), \qquad \angle(TG, BC) = \angle(BC, FA).$$

To make the resulting reflection explicit, scale coordinates so $F = (0, 0)$, $G = (1, 0)$, and $A = (u, v)$ with $v \neq 0$. The first angle identity makes the direction of FT proportional to $(1 - u, v)$, and the second makes the direction of GT proportional to $(-u, v)$. Their intersection is $T = (1 - u, v)$. Hence A and T have the same nonzero height above BC; since they are distinct, $AT \parallel BC$.

Certificate. Two reversed angle relations identify a reflection in the perpendicular bisector of FG. The coordinate finish records the correct side of the line.

G12. Close two circles around the target point

Source: IMO shortlist 2021, G1. **Record:** `imo_2021_18d688`.

Problem. In parallelogram $ABCD$, assume $AC = BC$. Take P beyond B on AB. Let $Q \neq D$ be the intersection of PD with (ACD), and $R \neq P$ the intersection of PC with (APQ). Prove CD, AQ, BR concurrent.

Solution. We have $AC = BC = AD$, so $\angle ABC = \angle BAC = \angle ACD = \angle ADC$. Using $(APQR)$ and $(AQCD)$ gives

$$\angle CRA = \angle CBA \pmod{180^\circ}.$$

Thus A, B, C, R lie on a circle. Set $X = AQ \cap CD$. Since $AB \parallel CD$, angles in $(APQR)$ give

$$\angle RQX = \angle RCX \pmod{180^\circ},$$

so C, Q, R, X are concyclic. Now use these two new circles and the original one:

$$\angle XRC = \angle XQC = -\angle CQA = \angle ADC = \angle BAC = -\angle CRB \pmod{180^\circ}.$$

It follows that X, R, B are collinear, so X is the desired concurrency point.

Certificate. Define the intended intersection first. Two auxiliary cyclicities force the last line through it.

G13. A circle equation linearizes the ratio

Source: IMO shortlist 2015, G4. **Record:** `imo_2015_b11e64`.

Problem. In an acute triangle ABC, let M be the midpoint of AC. A circle through B, M meets BA, BC again at P, Q. Complete parallelogram $BPTQ$, and assume $T \in (ABC)$. Determine BT/BM.

Solution. Put B at the vector origin, write $A = \mathbf{a}$, $C = \mathbf{c}$, $P = p\mathbf{a}$, $Q = q\mathbf{c}$, so $M = (\mathbf{a}+\mathbf{c})/2$ and $T = p\mathbf{a} + q\mathbf{c}$. Every circle through the origin has equation $|\mathbf{x}|^2 = \mathbf{u}\cdot\mathbf{x}$ for a fixed vector $\mathbf{u}$.

For (ABC), $\mathbf{u}\cdot\mathbf{a} = |\mathbf{a}|^2$ and $\mathbf{u}\cdot\mathbf{c} = |\mathbf{c}|^2$. Hence

$$BT^2 = p|\mathbf{a}|^2 + q|\mathbf{c}|^2.$$

For $(BPQM)$ the corresponding vector $\mathbf{v}$ satisfies $\mathbf{v}\cdot\mathbf{a} = p|\mathbf{a}|^2$ and $\mathbf{v}\cdot\mathbf{c} = q|\mathbf{c}|^2$. Therefore

$$BM^2 = \tfrac{1}{2}(p|\mathbf{a}|^2 + q|\mathbf{c}|^2) = BT^2/2.$$

The only possible ratio is $\sqrt{2}$.

It is attainable with all intersections interior to the sides. Take $BA = BC = 1$, $\cos B = 3/8$, and $p, q = (55 \pm \sqrt{385})/80$, both in $(0,1)$. Then $p+q = 11/8$ and $pq = 33/80$. Direct substitution gives $|M|^2 = 11/16 = (p+q)/2$ and $|T|^2 = (p+q)^2 - 2(1-3/8)pq = p+q$, exactly the two circle conditions. The isosceles triangle is acute. Thus the full answer is $\sqrt{2}$.

Certificate. The quadratic part of every circle equation is identical. Evaluating at a linear combination turns geometry into a linear identity.

G14. Transfer a tangent through equal powers

Source: IMO shortlist 2015, G6. **Record:** `imo_2015_d499b1`.

Problem. An acute triangle ABC has $AB > AC$, orthocenter H, base midpoint M, and altitude foot F on BC. On its circumcircle Ω, choose Q, K so that $\angle AQH = \angle QKH = 90^\circ$. Prove that (KQH) and (KFM) are tangent at K.

Solution. Let A' be the antipode of A on Ω, and let E be the reflection of H in BC. The standard orthocenter reflection argument puts E on Ω: reflection changes $\angle BHC = 180^\circ - A$ into the inscribed angle on the opposite side of BC. Also $BH \parallel CA'$ and $CH \parallel BA'$, so $BHCA'$ is a parallelogram and H, M, A' are collinear. Since $AQ \perp HQ$ and $AQ \perp A'Q$ (the angle in a semicircle), H, Q, A' are collinear. The lines HA' and HQ thus coincide.

Furthermore $EA' \parallel BC$, since $A' = 2M - H$ and E is the reflection of H in BC. Hence $\angle HEA' = 90^\circ$. The circles $\omega_1 = (KQH)$ and $\omega_2 = (EA'H)$ have diameters HQ and HA'; their common tangent at H, denoted t, is perpendicular to HM.

Let $R = QK \cap A'E$. These lines are the radical axes of Ω with ω_1, ω_2. Therefore R is on their mutual radical axis t. This intersection is finite: $A'E \parallel BC$, while $t \parallel BC$ would force $HM \perp BC$, contradicting $AB > AC$.

Since $\angle HKR = \angle HER = 90^\circ$, points H, K, E, R lie on the circle with diameter HR. Its center S, the midpoint of HR, satisfies $SK = SH = SE$. Thus S lies on the perpendicular bisector of HE, which is BC. It also lies on t.

The circle (HMF) has diameter HM, so its tangent at H is again t. Directed powers along BC give

$$\operatorname{Pow}_{(KFM)}(S) = \overline{SM}\,\overline{SF} = \operatorname{Pow}_{(HMF)}(S) = SH^2 = SK^2.$$

Since $K \in (KFM)$, this proves that SK is tangent to (KFM). Likewise $\operatorname{Pow}_{\omega_1}(S) = SH^2 = SK^2$, so SK is tangent to ω_1 at K. The two distinct circles share this tangent, proving the claim.

Certificate. A radical center manufactures a point with three equal powers. Tangency follows from the equality between power and the square of a known secant distance. The definitions of A', E and the collinearity argument are supplied here because the recovered source omitted that setup.

G15. Equal segments turn spiral similarity into rotation

Source: IMO shortlist 2024, G1. **Record:** `imo_2024_a60b6c`.

Problem. A cyclic quadrilateral satisfies $AC < BD < AD$ and $\angle DBA < 90^\circ$. On the side of AD opposite C, choose E with $DE \parallel AB$, $DE = AC$, and F with $AF \parallel CD$, $AF = BD$. Prove that the perpendicular bisectors of BC, EF meet on $(ABCD)$.

Solution. Cyclicity and the parallels give $\angle CAF = \angle EDB$ as directed angles, while $CA = DE$ and $AF = DB$. Thus ACF and DEB are congruent, so $CF = BE$. Put $P = BE \cap CF$. The same congruence gives

$$\angle CPB = \angle(CF, BE) = \angle(AF, DB) = \angle(DC, DB) = \angle CDB.$$

Consequently $P \in (ABCD)$, by the equal angles subtending CB.

Use the equal-segment center theorem proved in the companion theory volume: for two intersecting lines BE, CF, the direct similarity sending B to C and E to F has its center T on (PBC), allowing $T = P$. The proof solves $C - T = q(B - T)$ and $F - T = q(E - T)$ in complex coordinates and applies the cyclicity criterion, so it also covers coincident auxiliary intersections. Because $BE = CF$, the similarity has scale one, giving $TB = TC$ and $TE = TF$. Thus T belongs to both perpendicular bisectors. Since $(PBC) = (ABCD)$, it is their common point on the required circle.

Certificate. Prove equality of two segments, then use the center of the direct similarity. Equal scale produces both perpendicular bisectors simultaneously.

G16. Reflect a vertex in the incenter

Source: IMO shortlist 2024, G2. **Record:** `imo_2024_7db46b`.

Problem. A triangle has $AB < AC < BC$, incenter I, and incircle. The other tangent parallel to AC meets the interior of BC at X; the other tangent parallel to AB meets it at Y. Let $P \neq A$ be the intersection of AI with (ABC), and let K, L be the midpoints of AB, AC. Prove $\angle KIL + \angle YPX = 180^\circ$.

Solution. Reflect A in I to get A'. The half-turn about I preserves the incircle and sends AB, AC to their other parallel tangents. Therefore $A'X \parallel AC$ and $A'Y \parallel AB$. Moreover A, A', P are collinear. Angles in (ABC) now give

$$\angle APB = \angle ACB = \angle A'XC,$$

which proves B, P, A', X concyclic; similarly C, P, A', Y are concyclic. The homothety with center A and ratio two sends K, I, L to B, A', C, so $\angle KIL = \angle BA'C$.

The two auxiliary circles give $\angle APX = \angle A'BC$ and $\angle YPA = \angle BCA'$. In the given order these add to $\angle YPX$. Hence

$$\angle KIL + \angle YPX = \angle BA'C + \angle A'BC + \angle BCA' = 180^\circ.$$

Certificate. Reflecting a vertex in the incenter constructs both other parallel tangents at once. Doubling the midpoint triangle then reveals the ordinary triangle angle sum.

Attribution and proof audit

The printed records resolve to `https://mathnet.mit.edu/data_v2/full/imo/RECORD.json.gz`. The complete ledger credits original booklets and their editors. The official IMO 2017 paper is at https://www.imo-official.org/assets/documents/problems/2017/2017_eng.pdf . The rewritten expositions repair omitted definitions in G14, the angle-sum sign in G09, and missing branches in the G05 transcription. These repairs are recorded separately from the untouched, explicitly labeled source-supplied texts. A geometric diagram is never used as evidence of an unstated collinearity or point order.